



SCHOOL OF COMPUTER SCIENCES
UNIVERSITI SAINS MALAYSIA

CMT426/CMM426: BUSINESS INTELLIGENCE AND ANALYTICS
SEMESTER 2, ACADEMIC SESSION 2023/2024

GROUP 39

ASSIGNMENT 02

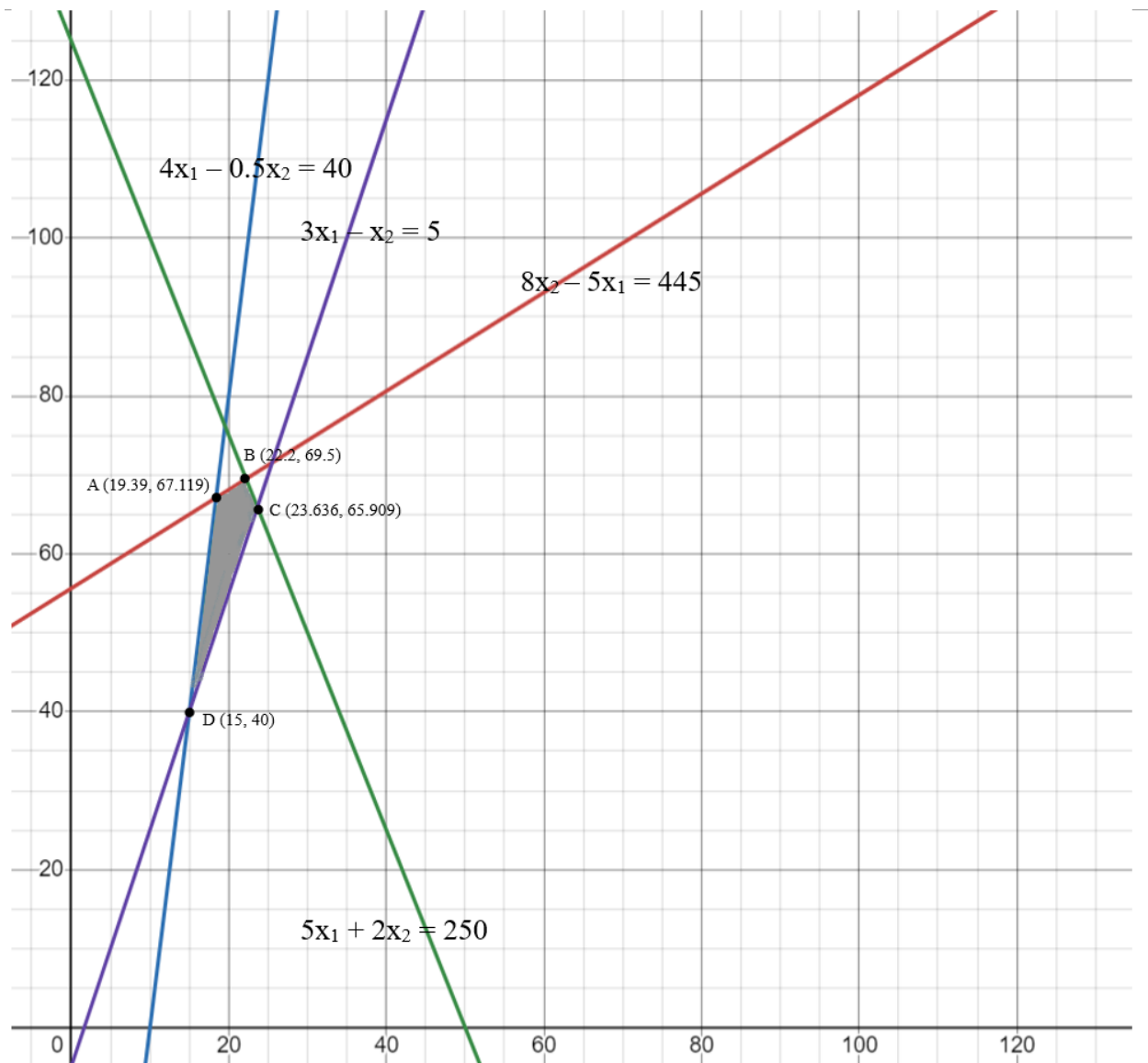
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QUESTION 1 (15 MARKS)

(a) Solve this model using graphical method.



Desmos, a graphical calculator applied to solve this linear programming problem. As a result, by fulfilling all resource constraints, a feasible region in grey colour was illustrated shown as above.

To identify the optimal solution, all coordinators will be inserted in the profit, $Z = 7x_1 + 5x_2$, as shown below:

Point A $(19.39, 67.119)$:

$$Z = 7x_1 + 5x_2$$

$$Z = 7(19.39) + 5(67.119)$$

$$Z = 135.73 + 335.595$$

$$Z = \mathbf{471.325}$$

Point B (22.2, 69.5):

$$Z = 7x_1 + 5x_2$$

$$Z = 7(22.2) + 5(69.5)$$

$$Z = 155.4 + 347.5$$

$$Z = \mathbf{502.9}$$

Point C (23.636, 65.909):

$$Z = 7x_1 + 5x_2$$

$$Z = 7(23.636) + 5(65.909)$$

$$Z = 165.452 + 329.545$$

$$Z = \mathbf{494.997}$$

Point D (15, 40):

$$Z = 7x_1 + 5x_2$$

$$Z = 7(15) + 5(40)$$

$$Z = 105 + 200$$

$$Z = \mathbf{305}$$

Thus, the optimal solution for x_1 and x_2 are 22.2 and 69.5 (Point B) as it could provide the maximum profit, which is 502.9 compared to other coordinators.

- (b) Use a software package based on the simplex method (e.g., Excel Solver) to solve the problem and then to generate sensitivity analysis. Explain the allowable range for each objective function coefficient, and the allowable range for each right-hand side. Give appropriate examples in your explanation.

Sensitivity report question 1:

Variable Cells

Cell	Name	Final Value	Reduced Cost	Objective Coefficient	Allowable Increase	Allowable Decrease
\$C\$12	Values x1	22.2	0	7	5.5	10.125
\$D\$12	Values x2	69.5	0	5	1E+30	2.2

Constraints

Cell	Name	Final Value	Shadow Price	Constraint R.H. Side	Allowable Increase	Allowable Decrease
\$E\$7	Resource 1 constraint Resource Used	445	0.22	445	66.9047619	167.2222222
\$E\$8	Resource 2 constraint Resource Used	54.05	0	40	14.05	1E+30
\$E\$9	Resource 3 constraint Resource Used	250	1.62	250	136.8181818	23.81355932
\$E\$10	Resource 4 constraint Resource Used	-25.1	0	5	1E+30	30.1

Allowable range for each objective function coefficient indicates the amount of increase or decrease the value would not change the optimal solution value, which is 22.2 and 69.5, as long as within the allowable range. For example,

The objective coefficient of **value x₁** is 7. It indicates to produce x₁, we need to spend 7 units of resources. Value x₁ has an allowable increase of 5.5 and an allowable decrease of 10.125. It indicates that as long as the expected profit of x₁ is in the range of -3.125 (= 7 - 10.125) and 12.5 (= 7 + 5.5), the final value of value x₁ (22.2) and x₂ (69.5) will remain the optimal quantities.

The objective coefficient of **value x₂** is 5. It indicates to produce x₂, we need to spend 5 units of resources. Value x₂ has an allowable increase of 1E+30 (known as infinite) and an allowable decrease of 2.2. It indicates that as long as the expected profit of x₂ is in the range of 2.8 (= 5 - 2.2) and infinite (= 5 + ∞), the final value of value x₁ (22.2) and x₂ (69.5) will remain the optimal quantities.

Next, in the constraints section, the shadow price provides the information of how much the optimal solution can be increased or decreased if the right-hand side (RHS) values changed by one unit. For example,

The shadow price and RHS value of resource 1 constraint are 0.22 and 445, respectively. The value of the allowance increase and allowable decrease indicates that the shadow price of 0.22 is valid for increase up to 66.9048 units of resources and decrease to 167.2222 units of resources. Therefore, the shadow price of 0.22 is applicable for increase up to 511.9048 units of resources (= 445 + 66.9048) and decrease down to 277.7778 units of resources (= 445 - 167.2222).

The shadow price and RHS value of resource 3 constraint are 1.62 and 250, respectively. The value of the allowance increase and allowable decrease indicates that the shadow price of 1.62 is valid for increase up to 136.8182 units of resources and decrease to 23.9136 units of resources. Therefore, the shadow price of 1.62 is applicable for increase up to 3868.182 units of resources ($= 250 + 136.8182$) and decrease down to 226.1865 units of resources ($= 250 - 23.8135$).

The shadow price and RHS value of resource 2 and resource 4 constraints used are 54.05 and 0, and -25.1 and 0, respectively. However, due to the resource 2 and resource 4 constraints are non-binding constraints. So, it is meaningless to increase the units of resources to 14.05 and decrease the units of resources to infinite for resource 2 constraints, and to increase the units of resources to infinite and decrease the units of resources to 30.1 for resource 4 constraint. It is because if both resources constraints are not fully utilised, producing extra units would not increase the total profit.

- (c) Based on the shadow prices obtained from Question 2(b), if you only can perform additional investment in one of the four resources, which resource that you want to invest in? Explain your answer using an appropriate example.

If I only can perform additional investment in one of the four resources, I would like to invest in resource 3. It is because its shadow price is 1.62, the highest shadow price compared to other resources. Thus, every extra resource produced will increase the profit of 1.62.

	A	B	C	D	E	F	G	H	I	J	K
1	LINEAR PROGRAMMING MODEL (QUESTION 1)										
2											
3			x1	x2							
4		Profit per decision variable	7	5							
5											
6		Resource	Requirement		Resource Used		Available Resource				
7		Resource 1 constraint	-5	8	445.0000	<=	445			Data cells	
8		Resource 2 constraint	4	-0.5	54.0500	>=	40			Changing cells	
9		Resource 3 constraint	5	2	250.0000	<=	250			Output cells	
10		Resource 4 constraint	2	-1	-25.1000	<=	5			Objective cell	
11							Total Profit				
12		Values	22.2000	69.5000			502.9000				
13											

Answer report Question 1

	A	B	C	D	E	F	G
14	Objective Cell (Max)						
15		Cell	Name	Original Value	Final Value		
16		\$G\$12	Values Total Profit	502.9000	502.9000		
17							
18							
19	Variable Cells						
20		Cell	Name	Original Value	Final Value	Integer	
21		\$C\$12	Values Requirement	22.2000	22.2000	Contin	
22		\$D\$12	Values x2	69.5000	69.5000	Contin	
23							
24							
25	Constraints						
26		Cell	Name	Cell Value	Formula	Status	Slack
27		\$E\$7	Resource 1 constraint Resource Used	445.0000	\$E\$7<=\$G\$7	Binding	0
28		\$E\$8	Resource 2 constraint Resource Used	54.0500	\$E\$8>=\$G\$8	Not Binding	14.0500
29		\$E\$9	Resource 3 constraint Resource Used	250.0000	\$E\$9<=\$G\$9	Binding	0
30		\$E\$10	Resource 4 constraint Resource Used	-25.1000	\$E\$10<=\$G\$10	Not Binding	30.1
31							

QUESTION 2 (15 MARKS)

Use a software package based on the simplex method (e.g., Excel Solver) to solve the problem and answer the following questions. All the values stated in the values must be rounded to 4 decimal places.

(a) What are the values for all the decision variables that maximize P.

	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
2																
3		A	B	C	D	E	F	G	H	I	J	K	L			
4	Profit per decision variable	365	365	365	400	400	400	360	360	360	290	290	290			
5																
6	Resource	Requirement												Resource Used		Available Resource
7	Constraint 1	1	0	0	-0.5	0	0	1	0	0	3	0	0	10.1924	<=	12
8	Constraint 2	0	2	0	0	1	0	0	1	0	0	1	0	19.0937	>=	18
9	Constraint 3	0	0	1	0	0	2	0	0	1	0	0	4	10.0000	<=	10
10	Constraint 4	2	3	1	0	0	0	0	0	0	0	0	0	20.5510	>=	20
11	Constraint 5	0	0	0	1	1	0.25	0	0	0	0	0	0	6.0000	>=	6
12	Constraint 6	0	0	0	0	0	0	1	3	1	0	0	0	50.0000	>=	50
13	Constraint 7	0	0	0	0	0	0	0	0	0	-0.8	1	-0.3	-0.7485	<=	5
14	Constraint 8	500	0	0	700	0	0	600	0	0	400	0	0	6400.0000	<=	6400
15	Constraint 9	0	250	0	0	520	0	0	300	0	0	200	0	6120.0000	<=	6120
16	Constraint 10	0	0	550	0	0	450	0	0	200	0	0	100	2000.0000	<=	2100
17	Constraint 11	0.0833	-0.0556	0	0.0833	-0.0556	0	0.0833	-0.0556	0	0.0833	-0.0556	0	0.0000	"_B"	0
18	Constraint 12	0.0833	0	-0.1	0.0833	0	-0.1	0.0833	0	-0.1	0.0833	0	-0.1	0.0000	"_B"	0
19																Total Profit
20	Values	8.6134	1.1081	0.0000	2.4558	3.5442	0.0000	0.0000	13.3333	10.0000	0.9356	0.0000	0.0000			14619.6615
21																
22																

By using Microsoft Excel Solver, the values for all the decision variables that maximize P are shown as below:

Decision variable A = 8.6134

Decision variable B = 1.1081

Decision variable C = 0

Decision variable D = 2.4558

Decision variable E = 3.5442

Decision variable F = 0

Decision variable G = 0

Decision variable H = 13.3333

Decision variable I = 10.0000

Decision variable J = 0.9356

Decision variable K = 0

Decision variable L = 0

P = **14619.6615**

- (b) Identify the non-binding constraints. For each of the identified non-binding constraints, explain the slack/surplus value.

	A	B	C	D	E	F	G
34							
35			Constraints				
36			Cell	Name	Cell Value	Formula	Status
37			\$O\$10	Constraint 4 Resource Used	20.55097978	\$O\$10>=\$Q\$10	Not Binding
38			\$O\$11	Constraint 5 Resource Used	6	\$O\$11>=\$Q\$11	Binding
39			\$O\$12	Constraint 6 Resource Used	50	\$O\$12>=\$Q\$12	Binding
40			\$O\$13	Constraint 7 Resource Used	-0.748508687	\$O\$13<=\$Q\$13	Not Binding
41			\$O\$14	Constraint 8 Resource Used	6400	\$O\$14<=\$Q\$14	Binding
42			\$O\$15	Constraint 9 Resource Used	6120	\$O\$15<=\$Q\$15	Binding
43			\$O\$16	Constraint 10 Resource Used	2000	\$O\$16<=\$Q\$16	Not Binding
44			\$O\$17	Constraint 11 Resource Used	9.71445E-17	\$O\$17=\$Q\$17	Binding
45			\$O\$18	Constraint 12 Resource Used	-4.57967E-16	\$O\$18=\$Q\$18	Binding
46			\$O\$7	Constraint 1 Resource Used	10.19235395	\$O\$7<=\$Q\$7	Not Binding
47			\$O\$8	Constraint 2 Resource Used	19.09370282	\$O\$8>=\$Q\$8	Not Binding
48			\$O\$9	Constraint 3 Resource Used	10	\$O\$9<=\$Q\$9	Binding
49							

In the Answer Report Q2 sheet of Microsoft Excel Solver, the constraint that identified as non-binding constraints are **constraint 1, constraint 2, constraint 4, constraint 7, and constraint 10**.

Constraint 1 is a non-binding constraint because it has a slack value of 1.8076 ($= 12 - 10.1924$). It indicates that constraint 1 has a total of 1.8076 resources that unused for certain purposes as the maximum requirement is 12

Constraint 2 is a non-binding constraint because it has a surplus value of 1.0937 ($= 19.0937 - 18$). It indicates that constraint 2 has exceed a total of 1.0937 resources that used for certain purposes as the minimum requirement is 18.

Constraint 4 is a non-binding constraint because it has a surplus value of 0.5510 ($= 20.5510 - 20$). It indicates that constraint 4 has exceed a total of 0.5510 resources that used for certain purposes as the minimum requirement is 20.

Constraint 7 is a non-binding constraint because it has a slack value of 5.7485 [$= 5 - (-0.7485)$]. It indicates that constraint 7 has a total of 5.7485 resources that unused for certain purposes as the maximum requirement is 5.

Constraint 10 is a non-binding constraint because it has a slack value of 100 ($= 2100 - 2000$). It indicates that constraint 10 has a total of 100 resources that unused for certain purposes as the maximum requirement is 2100.

Answer report question 2:

	A	B	C	D	E	F	G
14	Objective Cell (Max)						
15		Cell	Name	Original Value	Final Value		
16		\$Q\$20	Values Total Profit	14619.66155	14619.66155		
17							
18							
19	Variable Cells						
20		Cell	Name	Original Value	Final Value	Integer	
21		\$C\$20	Values A	8.613352935	8.613352935	Contin	
22		\$D\$20	Values B	1.108091305	1.108091305	Contin	
23		\$E\$20	Values C	0	0	Contin	
24		\$F\$20	Values D	2.455813127	2.455813127	Contin	
25		\$G\$20	Values E	3.544186873	3.544186873	Contin	
26		\$H\$20	Values F	0	0	Contin	
27		\$I\$20	Values G	0	0	Contin	
28		\$J\$20	Values H	13.33333333	13.33333333	Contin	
29		\$K\$20	Values I	10	10	Contin	
30		\$L\$20	Values J	0.935635858	0.935635858	Contin	
31		\$M\$20	Values K	0	0	Contin	
32		\$N\$20	Values L	0	0	Contin	
33							
34							
35	Constraints						
36		Cell	Name	Cell Value	Formula	Status	Slack
37		\$O\$10	Constraint 4 Resource Used	20.55097978	\$O\$10>=\$Q\$10	Not Binding	0.550979784
38		\$O\$11	Constraint 5 Resource Used	6	\$O\$11>=\$Q\$11	Binding	0
39		\$O\$12	Constraint 6 Resource Used	50	\$O\$12>=\$Q\$12	Binding	0
40		\$O\$13	Constraint 7 Resource Used	-0.748508687	\$O\$13<=\$Q\$13	Not Binding	5.748508687
41		\$O\$14	Constraint 8 Resource Used	6400	\$O\$14<=\$Q\$14	Binding	0
42		\$O\$15	Constraint 9 Resource Used	6120	\$O\$15<=\$Q\$15	Binding	0
43		\$O\$16	Constraint 10 Resource Used	2000	\$O\$16<=\$Q\$16	Not Binding	100
44		\$O\$17	Constraint 11 Resource Used	9.71445E-17	\$O\$17=\$Q\$17	Binding	0
45		\$O\$18	Constraint 12 Resource Used	-4.57967E-16	\$O\$18=\$Q\$18	Binding	0
46		\$O\$7	Constraint 1 Resource Used	10.19235395	\$O\$7<=\$Q\$7	Not Binding	1.807646053
47		\$O\$8	Constraint 2 Resource Used	19.09370282	\$O\$8>=\$Q\$8	Not Binding	1.093702816
48		\$O\$9	Constraint 3 Resource Used	10	\$O\$9<=\$Q\$9	Binding	0

Sensitivity report for question 2:

	A	B	C	D	E	F	G	H
6	Variable Cells							
7				Final	Reduced	Objective	Allowable	Allowable
8	Cell	Name		Value	Cost	Coefficient	Increase	Decrease
9	\$C\$20	Values A		8.613352935	0	365	69.54242034	32
10	\$D\$20	Values B		1.108091305	0	365	150	21.81818182
11	\$E\$20	Values C		0	-5.925925926	365	5.925925926	1E+30
12	\$F\$20	Values D		2.455813127	0	400	96	255
13	\$G\$20	Values E		3.544186873	0	400	255	96
14	\$H\$20	Values F		0	-707.7717102	400	707.7717102	1E+30
15	\$I\$20	Values G		0	-69.07407407	360	69.07407407	1E+30
16	\$J\$20	Values H		13.33333333	0	360	17.77777778	207.2222222
17	\$K\$20	Values I		10	0	360	1E+30	5.925925926
18	\$L\$20	Values J		0.935635858	0	290	48	73.83269133
19	\$M\$20	Values K		0	-47.22222222	290	47.22222222	1E+30
20	\$N\$20	Values L		0	-2377.713279	290	2377.713279	1E+30
21								
22	Constraints							
23				Final	Shadow	Constraint	Allowable	Allowable
24	Cell	Name		Value	Price	R.H. Side	Increase	Decrease
25	\$O\$10	Constraint 4 Resource Used		20.55097978	0	20	0.550979784	1E+30
26	\$O\$11	Constraint 5 Resource Used		6	-115	6	0.091829964	0.467817929
27	\$O\$12	Constraint 6 Resource Used		50	-10.92592593	50	0.464889193	7.578650453
28	\$O\$13	Constraint 7 Resource Used		-0.748508687	0	5	1E+30	5.748508687
29	\$O\$14	Constraint 8 Resource Used		6400	0.75	6400	93.56358584	27.54898922
30	\$O\$15	Constraint 9 Resource Used		6120	0.555555556	6120	126.3108409	49.5881806
31	\$O\$16	Constraint 10 Resource Used		2000	0	2100	1E+30	100
32	\$O\$17	Constraint 11 Resource Used		9.71445E-17	-4066.746603	0	0.020783122	0.043418212
33	\$O\$18	Constraint 12 Resource Used		-4.57967E-16	3946.698584	0	0.005986203	0.010024953
34	\$O\$7	Constraint 1 Resource Used		10.19235395	0	12	1E+30	1.807646053
35	\$O\$8	Constraint 2 Resource Used		19.09370282	0	18	1.093702816	1E+30
36	\$O\$9	Constraint 3 Resource Used		10	765.5957843	10	0.068709493	0.101593396

QUESTION 3 (20 MARKS)

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	AA	AB
1		The Company 99 Speed Courier Product-Mix Problem																										
2																												
3		Route	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20						
4		Time (in hours)	6	8	6	6	7	5	7	8	9	9	8	9	6	5	7	5	5	8	7	5						
5																												
6		Delivery Locations	Delivery route																					Route delivery needed	Available route delivery			
7		Location 1	1	1	0	1	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	1	"=	1			Data cells
8		Location 2	0	0	0	0	0	0	1	1	0	0	0	0	0	0	0	0	1	0	1	0	1	"=	1			Changing cells
9		Location 3	0	0	1	0	0	0	0	0	0	1	0	0	0	0	1	0	0	0	0	0	1	"=	1			Output cells
10		Location 4	0	1	0	0	0	1	0	0	1	0	0	0	0	0	0	0	1	1	0	0	1	"=	1			Objective cell
11		Location 5	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	1	"=	1			
12		Location 6	1	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	1	"=	1			
13		Location 7	0	0	0	0	0	1	0	0	0	0	0	0	1	0	0	0	0	0	0	0	1	"=	1			
14		Location 8	0	0	1	0	0	0	0	0	0	1	0	0	0	0	0	1	0	0	0	0	1	"=	1			
15		Location 9	0	0	1	0	0	0	0	0	1	0	0	0	0	0	1	0	0	0	0	0	1	"=	1			
16		Location 10	0	0	0	0	0	1	0	0	0	0	0	0	0	1	0	0	0	1	0	0	1	"=	1			
17		Location 11	0	0	1	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	1	0	1	"=	1		
18		Location 12	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	1	"=	1		
19		Location 13	0	0	0	0	0	1	0	0	1	0	0	0	1	0	0	0	0	0	0	0	0	1	"=	1		
20		Location 14	0	0	0	1	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	1	"=	1			
21		Location 15	0	0	0	0	1	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	1	"=	1			
22		Location 16	0	0	0	0	1	0	1	0	0	0	0	0	0	0	0	0	0	1	0	0	1	"=	1			
23		Location 17	0	1	0	0	0	0	0	0	0	0	0	0	1	0	0	1	0	0	0	0	1	"=	1			
24		Location 18	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	1	0	0	1	"=	1			
25		Location 19	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	1	1	0	0	0	1	"=	1			
26		Location 20	0	0	0	0	1	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	1	"=	1			
27																												
28			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20			Minimum time			
29		Routes used	0	0	1	0	1	0	0	0	0	0	1	1	1	1	0	0	1	0	0	0			46			
30																												

By using Microsoft Excel Solver, the objective is to help Ali to determine the minimise number of trucks needed which implied routes that minimise the total traveling time.

Decision Variable

The decision variable is the cell C7:V26 which are all the routes. The combine two elements together will become objective function.

Result Variable

The result variable is the total number of trucks and total traveling time.

Objective Function

The equation of the objective function is Min Total traveling = $6R_1 + 8R_2 + 6R_3 + 6R_4 + 7R_5 + 5R_6 + 7R_7 + 8R_8 + 9R_9 + 9R_{10} + 8R_{11} + 9R_{12} + 6R_{13} + 5R_{14} + 7R_{15} + 5R_{16} + 5R_{17} + 8R_{18} + 7R_{19} + 5R_{20}$

Constraints

Constraint 1 is Location 1 = $R_1 + R_2 + R_4 + R_{11} = 1$

Constraint 2 is Location 2 = $R_7 + R_8 + R_{17} + R_{19} = 1$

Constraint 3 is Location 3 = $R_3 + R_{10} + R_{15}$

Constraint 4 is Location 4 = $R_2 + R_6 + R_9 + R_{17} + R_{18}$

Constraint 5 is Location 5 = R_{12}

Constraint 6 is Location 6 = $R_1 + R_{11}$

Constraint 7 is Location 7 = $R_6 + R_{12} + R_{20}$

Constraint 8 is Location 8 = $R_3 + R_{10} + R_{16}$

Constraint 9 is Location 9 = $R_3 + R_9 + R_{15}$

Constraint 10 is Location 10 = $R_6 + R_{14} + R_{18}$

Constraint 11 is Location 11 = $R_3 + R_8 + R_{19}$

Constraint 12 is Location 12 = R_{11}

Constraint 13 is Location 13 = $R_6 + R_9 + R_{13}$

Constraint 14 is Location 14 = $R_4 + R_{12} + R_{20}$

Constraint 15 is Location 15 = $R_5 + R_8$

Constraint 16 is Location 16 = $R_5 + R_7 + R_{18}$

Constraint 17 is Location 17 = $R_2 + R_{13} + R_{16}$

Constraint 18 is Location 18 = $R_{14} + R_{18}$

Constraint 19 is Location 19 = $R_4 + R_{16} + R_{17}$

Constraint 20 is Location 20 = $R_5 + R_{15}$

Optimal Solution

The optimal solution for the route used are **Route 3, Route 5, Route 11, Route 12, Route 13, Route 14 and Route 17.**

The delivery location for route 3 are Location 3, Location 8, Location 9 and Location 11.

The delivery location for route 5 is Location 15, Location 16 and Location 20.

The delivery location for route 11 is Location 1, Location 6 and Location 12.

The delivery location for route 12 is Location 5, Location 7 and Location 14.

The delivery location for route 13 is Location 13 and Location 17.

The delivery location for route 14 is Location 10 and Location 18.

The delivery location for route 17 is Location 2, Location 4 and Location 19.

The minimum traveling time required 46 hours.

Answer report question 3:

Objective Cell (Min)

Cell	Name	Original Value	Final Value
\$Y\$29	Routes used Minimum time	0	46

Variable Cells

Cell	Name	Original Value	Final Value	Integer
\$C\$29	Routes used Delivery route	0	0	Contin
\$D\$29	Routes used	0	0	Contin
\$E\$29	Routes used	0	1	Contin
\$F\$29	Routes used	0	0	Contin
\$G\$29	Routes used	0	1	Contin
\$H\$29	Routes used	0	0	Contin
\$I\$29	Routes used	0	0	Contin
\$J\$29	Routes used	0	0	Contin
\$K\$29	Routes used	0	0	Contin
\$L\$29	Routes used	0	0	Contin
\$M\$29	Routes used	0	1	Contin
\$N\$29	Routes used	0	1	Contin
\$O\$29	Routes used	0	1	Contin
\$P\$29	Routes used	0	1	Contin
\$Q\$29	Routes used	0	0	Contin
\$R\$29	Routes used	0	0	Contin
\$S\$29	Routes used	0	1	Contin
\$T\$29	Routes used	0	0	Contin
\$U\$29	Routes used	0	0	Contin
\$V\$29	Routes used	0	0	Contin

Constraints

Cell	Name	Cell Value	Formula	Status	Slack
\$W\$7	Location 1 Route delivery needed	1	\$W\$7=\$Y\$7	Binding	0
\$W\$8	Location 2 Route delivery needed	1	\$W\$8=\$Y\$8	Binding	0
\$W\$9	Location 3 Route delivery needed	1	\$W\$9=\$Y\$9	Binding	0
\$W\$10	Location 4 Route delivery needed	1	\$W\$10=\$Y\$10	Binding	0
\$W\$11	Location 5 Route delivery needed	1	\$W\$11=\$Y\$11	Binding	0
\$W\$12	Location 6 Route delivery needed	1	\$W\$12=\$Y\$12	Binding	0
\$W\$13	Location 7 Route delivery needed	1	\$W\$13=\$Y\$13	Binding	0
\$W\$14	Location 8 Route delivery needed	1	\$W\$14=\$Y\$14	Binding	0
\$W\$15	Location 9 Route delivery needed	1	\$W\$15=\$Y\$15	Binding	0
\$W\$16	Location 10 Route delivery needed	1	\$W\$16=\$Y\$16	Binding	0
\$W\$17	Location 11 Route delivery needed	1	\$W\$17=\$Y\$17	Binding	0
\$W\$18	Location 12 Route delivery needed	1	\$W\$18=\$Y\$18	Binding	0
\$W\$19	Location 13 Route delivery needed	1	\$W\$19=\$Y\$19	Binding	0
\$W\$20	Location 14 Route delivery needed	1	\$W\$20=\$Y\$20	Binding	0
\$W\$21	Location 15 Route delivery needed	1	\$W\$21=\$Y\$21	Binding	0
\$W\$22	Location 16 Route delivery needed	1	\$W\$22=\$Y\$22	Binding	0
\$W\$23	Location 17 Route delivery needed	1	\$W\$23=\$Y\$23	Binding	0
\$W\$24	Location 18 Route delivery needed	1	\$W\$24=\$Y\$24	Binding	0
\$W\$25	Location 19 Route delivery needed	1	\$W\$25=\$Y\$25	Binding	0
\$W\$26	Location 20 Route delivery needed	1	\$W\$26=\$Y\$26	Binding	0

In the Answer Report Q3 sheet of Microsoft Excel Solver, all the binding constraints and there no slack.

Sensitivity report for report Q3:

	A	B	C	D	E	F	G	H
6	Variable Cells							
7				Final	Reduced	Objective	Allowable	Allowable
8	Cell	Name		Value	Cost	Coefficient	Increase	Decrease
9	\$C\$29	Routes used Delivery route		0	0	6	1E+30	1E+30
10	\$D\$29	Routes used		0	0	8	1E+30	1E+30
11	\$E\$29	Routes used		1	0	6	1E+30	1E+30
12	\$F\$29	Routes used		0	0	6	1E+30	1E+30
13	\$G\$29	Routes used		1	0	7	1E+30	1E+30
14	\$H\$29	Routes used		0	0	5	1E+30	1E+30
15	\$I\$29	Routes used		0	0	7	1E+30	1E+30
16	\$J\$29	Routes used		0	0	8	1E+30	1E+30
17	\$K\$29	Routes used		0	0	9	1E+30	1E+30
18	\$L\$29	Routes used		0	0	9	1E+30	1E+30
19	\$M\$29	Routes used		1	0	8	1E+30	1E+30
20	\$N\$29	Routes used		1	0	9	1E+30	1E+30
21	\$O\$29	Routes used		1	0	6	1E+30	1E+30
22	\$P\$29	Routes used		1	0	5	1E+30	1E+30
23	\$Q\$29	Routes used		0	0	7	1E+30	1E+30
24	\$R\$29	Routes used		0	0	5	1E+30	1E+30
25	\$S\$29	Routes used		1	0	5	1E+30	1E+30
26	\$T\$29	Routes used		0	0	8	1E+30	1E+30
27	\$U\$29	Routes used		0	0	7	1E+30	1E+30
28	\$V\$29	Routes used		0	0	5	1E+30	1E+30

	A	B	C	D	E	F	G	H
30	Constraints							
31				Final	Shadow	Constraint	Allowable	Allowable
32	Cell	Name		Value	Price	R.H. Side	Increase	Decrease
33	\$W\$7	Location 1 Route delivery needed		1	-4	1	0	0
34	\$W\$8	Location 2 Route delivery needed		1	11	1	1	0
35	\$W\$9	Location 3 Route delivery needed		1	-4	1	0	1
36	\$W\$10	Location 4 Route delivery needed		1	7	1	0	0
37	\$W\$11	Location 5 Route delivery needed		1	4	1	0	1
38	\$W\$12	Location 6 Route delivery needed		1	10	1	0	0
39	\$W\$13	Location 7 Route delivery needed		1	-18	1	0	0
40	\$W\$14	Location 8 Route delivery needed		1	13	1	1	0
41	\$W\$15	Location 9 Route delivery needed		1	1	1	0	0
42	\$W\$16	Location 10 Route delivery needed		1	15	1	0	0
43	\$W\$17	Location 11 Route delivery needed		1	-4	1	0	0
44	\$W\$18	Location 12 Route delivery needed		1	2	1	0	1
45	\$W\$19	Location 13 Route delivery needed		1	1	1	0	1
46	\$W\$20	Location 14 Route delivery needed		1	23	1	0	0
47	\$W\$21	Location 15 Route delivery needed		1	1	1	0	0
48	\$W\$22	Location 16 Route delivery needed		1	-4	1	0	0
49	\$W\$23	Location 17 Route delivery needed		1	5	1	1	0
50	\$W\$24	Location 18 Route delivery needed		1	-10	1	0	0
51	\$W\$25	Location 19 Route delivery needed		1	-13	1	0	0.5
52	\$W\$26	Location 20 Route delivery needed		1	10	1	0	0

QUESTION 4 (20 MARKS)

a) Formulate a linear programming model

Objective: Minimize the total cost of planting

Let:

- x_{11} = Acres of Plant A in Farm 1
- x_{12} = Acres of Plant A in Farm 2
- x_{13} = Acres of Plant A in Farm 3
- x_{21} = Acres of Plant B in Farm 1
- x_{22} = Acres of Plant B in Farm 2
- x_{23} = Acres of Plant B in Farm 3
- x_{31} = Acres of Plant C in Farm 1
- x_{32} = Acres of Plant C in Farm 2
- x_{33} = Acres of Plant C in Farm 3

Objective function:

$$\text{Minimize } Z = 90x_{11} + 50x_{12} + 45x_{13} + 75x_{21} + 80x_{22} + 50x_{23} + 25x_{31} + 50x_{32} + 90x_{33}$$

Constraints:

- Yield constraints:

$$420x_{11} + 300x_{12} + 500x_{13} \geq 21000$$

$$200x_{21} + 250x_{22} + 250x_{23} \geq 25000$$

$$300x_{31} + 150x_{32} + 125x_{33} \geq 30000$$

- Acre constraints:

$$x_{11} + x_{21} + x_{31} \leq 100$$

$$x_{12} + x_{22} + x_{32} \leq 50$$

$$x_{13} + x_{23} + x_{33} \leq 100$$

- Non-negativity constraints:

$$x_{ij} \geq 0 \text{ for all } i \text{ and } j$$

b) Results from Excel Solver

	A	B	C	D	E	F	G	H	I	J	K
1	Question 4 (Ali's Farm)										
2											
3		Cost/acre of plant (RM)	Plant A	Plant B	Plant C						
4		Farm 1	90	75	25						
5		Farm 2	50	80	50						
6		Farm 3	45	50	90						
7											
8		Plant yield/acre (kg)	Plant A	Plant B	Plant C						
9		Farm 1	420	200	300						Data cells
10		Farm 2	300	250	150						Changing cells
11		Farm 3	500	250	125						Output cells
12											Objective cell
13		Batches produced	Plant A	Plant B	Plant C	Acres Used		Maximum Acres			
14		Farm 1	0	0	100	100	<=	100			
15		Farm 2	0	42	0	42	<=	50			
16		Farm 3	42	58	0	100	<=	100			
17		Yield Used	21000	25000	30000						
18			>=	>=	>=			Total Cost (RM)			
19		Required Yield (kg)	21000	25000	30000			10650			
20											

Changing variable cells:

- $x_{11} = 0$ acres
- $x_{12} = 0$ acres
- $x_{13} = 42$ acres
- $x_{21} = 0$ acres
- $x_{22} = 42$ acres
- $x_{23} = 58$ acres
- $x_{31} = 100$ acres
- $x_{32} = 0$ acres
- $x_{33} = 0$ acres

Based on the results in Excel Solver to minimize the cost of planting plan to meet the demands, 42 acres of Farm 3 is used to produce Plant A, 42 acres of Farm 2 and 58 acres of Farm 3 is used to produce Plant B, and 100 acres of Farm 1 is used to produce Plant C.

Objective Cell:

$$H19 = C14 \cdot C4 + D14 \cdot D4 + E14 \cdot E4 + C15 \cdot C5 + D15 \cdot D5 + E15 \cdot E5 + C16 \cdot C6 + D16 \cdot D6 + E16 \cdot E6 = \text{RM } 10650$$

Based on the acres used for each plant, we can minimize the cost to RM 10, 650.

Output Cells:

- Yield used:

$$C14 \cdot C9 + C15 \cdot C10 + C16 \cdot C11 \geq 21000 = \text{21000 kg}$$

$$D14 \cdot D9 + D15 \cdot D10 + D16 \cdot D11 \geq 25000 = \text{25000 kg}$$

$$E14 \cdot E9 + E15 \cdot E10 + E16 \cdot E11 \geq 30000 = \mathbf{30000 \text{ kg}}$$

- Acres used:

$$C14 + D14 + E14 \leq 100 = \mathbf{100 \text{ acres}}$$

$$C15 + D15 + E15 \leq 50 = \mathbf{42 \text{ acres}}$$

$$C16 + D16 + E16 \leq 100 = \mathbf{100 \text{ acres}}$$

To produce the required yield of 21000kg of Plant A, 25000kg of plant B, and 30000kg of plant C, a total of 100 acres of Farm 1, 42 acres of Farm 2 and 100 acres of Farm 3 is used to meet the demands.

c) Is expanding the area of Farm 1 a cost-saving idea?

No, expanding the area of Farm 1 is not a cost-saving idea. The shadow price of 0 for Farm 1 acres indicates that increasing the available land in Farm 1 will have no effect on the total cost.

Cell	Name	Final Value	Shadow Price
\$C\$17	Yield Used Plant A	21000	0.15
\$D\$17	Yield Used Plant B	25000	0.32
\$E\$17	Yield Used Plant C	30000	0.083333333
\$F\$14	Farm 1 Acres Used	100	0
\$F\$15	Farm 2 Acres Used	42	0
\$F\$16	Farm 3 Acres Used	100	-30

With the current setup and shadow prices provided:

- It is more beneficial to focus on the utilization of Farm 3, where reducing the area might decrease the total cost due to the negative shadow price.
- Plant yields have positive shadow prices, indicating that producing additional yields will increase the total cost.

d) Analyze the consequence of fully utilizing all the three farms to meet the demands. Is this a wise decision to do so?

Batches produced	Plant A	Plant B	Plant C	Acres Used		Maximum Acres
Farm 1	0	0	100	100	<=	100
Farm 2	20	30	0	50	<=	50
Farm 3	30	70	0	100	<=	100
Yield Used	21000	25000	30000			
	>=	>=	>=			Total Cost (RM)
Required Yield (kg)	21000	25000	30000			10750

Fully utilizing all three farms is not a wise decision. This is because:

- The optimized solution already meets the demand at a lower cost.
- Using all available land increases the total cost by RM 100 without any additional benefits.
- Efficiency in resource allocation is crucial, and the optimized solution demonstrates better resource utilization.

It's best that Ali sticks to the optimized planting plan provided by the Solver, as it is the most cost-effective way to meet the demands. Fully utilizing all the land results in unnecessary additional costs.

Answer report question 4:

	A	B	C	D	E	F	G
14	Objective Cell (Min)						
15	Cell		Name	Original Value	Final Value		
16	\$H\$19		Required Yield (kg) Total Cost (RM)	10650	10650		
17							
18							
19	Variable Cells						
20	Cell		Name	Original Value	Final Value	Integer	
21	\$C\$14		Farm 1 Plant A	0	0	Contin	
22	\$D\$14		Farm 1 Plant B	0	0	Contin	
23	\$E\$14		Farm 1 Plant C	100	100	Contin	
24	\$C\$15		Farm 2 Plant A	0	0	Contin	
25	\$D\$15		Farm 2 Plant B	42	42	Contin	
26	\$E\$15		Farm 2 Plant C	0	0	Contin	
27	\$C\$16		Farm 3 Plant A	42	42	Contin	
28	\$D\$16		Farm 3 Plant B	58	58	Contin	
29	\$E\$16		Farm 3 Plant C	0	0	Contin	
30							
31							
32	Constraints						
33	Cell		Name	Cell Value	Formula	Status	Slack
34	\$C\$17		Yield Used Plant A	21000	\$C\$17>=\$C\$19	Binding	0
35	\$D\$17		Yield Used Plant B	25000	\$D\$17>=\$D\$19	Binding	0
36	\$E\$17		Yield Used Plant C	30000	\$E\$17>=\$E\$19	Binding	0
37	\$F\$14		Farm 1 Acres Used	100	\$F\$14<=\$H\$14	Binding	0
38	\$F\$15		Farm 2 Acres Used	42	\$F\$15<=\$H\$15	Not Binding	8
39	\$F\$16		Farm 3 Acres Used	100	\$F\$16<=\$H\$16	Binding	0

Sensitivity report question 4:

	A	B	C	D	E	F	G	H
6	Variable Cells							
7				Final	Reduced	Objective	Allowable	Allowable
8	Cell	Name	Value	Cost	Coefficient	Increase	Decrease	
9	\$C\$14	Farm 1 Plant A	0	27	90	1E+30	27	
10	\$D\$14	Farm 1 Plant B	0	11	75	1E+30	11	
11	\$E\$14	Farm 1 Plant C	100	0	25	75	25	
12	\$C\$15	Farm 2 Plant A	0	5	50	1E+30	5	
13	\$D\$15	Farm 2 Plant B	42	0	80	8.333333333	30	
14	\$E\$15	Farm 2 Plant C	0	37.5	50	1E+30	37.5	
15	\$C\$16	Farm 3 Plant A	42	0	45	8.333333333	75	
16	\$D\$16	Farm 3 Plant B	58	0	50	30	8.333333333	
17	\$E\$16	Farm 3 Plant C	0	109.5833333	90	1E+30	109.5833333	
18								
19	Constraints							
20			Final	Shadow	Constraint	Allowable	Allowable	
21	Cell	Name	Value	Price	R.H. Side	Increase	Decrease	
22	\$C\$17	Yield Used Plant A	21000	0.15	21000	4000	21000	
23	\$D\$17	Yield Used Plant B	25000	0.32	25000	2000	10500	
24	\$E\$17	Yield Used Plant C	30000	0.083333333	30000	0	30000	
25	\$F\$14	Farm 1 Acres Used	100	0	100	1E+30	0	
26	\$F\$15	Farm 2 Acres Used	42	0	50	1E+30	8	
27	\$F\$16	Farm 3 Acres Used	100	-30	100	42	8	

QUESTION 5 (30 MARKS)

(a) 3 workers in a group, 7 groups of workers, each group with 3 workers except one group with 2 workers.

Objective: Minimize makespan

Constraints: Each job assigned to one group, total processing time per group $\leq$ available hours

Generated schedule and analysis:

Group 1:

Order	Processing Time (hours)	Flow Time (hours)	Due Date (-th hour)	Job Lateness
G1	18	6	≤ 29	0
G1	20	8	≤ 33	0
G1	8	7	≤ 26	0

Group 2:

Order	Processing Time (hours)	Flow Time (hours)	Due Date (-th hour)	Job Lateness
G2	13	8	≤ 23	0
G2	10	7	≤ 17	0
G2	5	7	≤ 22	0

Group 3:

Order	Processing Time (hours)	Flow Time (hours)	Due Date (-th hour)	Job Lateness
G3	7	8	≤ 17	0
G3	2	9	≤ 24	0
G3	4	5	≤ 36	0

Group 4:

Order	Processing Time (hours)	Flow Time (hours)	Due Date (-th hour)	Job Lateness
G4	3	5	≤ 15	0

G4	14	3	≤ 10	0
G4	19	12	≤ 34	0

Group 5:

Order	Processing Time (hours)	Flow Time (hours)	Due Date (-th hour)	Job Lateness
G5	9	4	≤ 20	0
G5	12	5	≤ 14	0
G5	1	13	≤ 25	0

Group 6:

Order	Processing Time (hours)	Flow Time (hours)	Due Date (-th hour)	Job Lateness
G6	16	2	≤ 8	0
G6	17	7	≤ 16	0
G6	11	11	≤ 29	0

Group 7:

Order	Processing Time (hours)	Flow Time (hours)	Due Date (-th hour)	Job Lateness
G7	6	8	≤ 18	0
G7	15	12	≤ 25	0

Parameters :

Order: The sequence number of the job.

Processing Time (hours): The time required to complete each job.

Due Date (-th hour): The deadline by which the job must be completed to avoid lateness.

Group Setting: The assignment of jobs into groups. Here, we have 7 groups with 3 workers each, except one group with 2 workers.

Order (within groups): The specific sequence in which jobs are processed within each group.

Flow Time (hours): The cumulative time taken for each job to be completed, considering its start and end within the group schedule.

Due Date (within groups): The respective due date for jobs within their groups.

Job Lateness: The difference between the due date and the flow time. If the flow time exceeds the due date, it indicates job lateness, otherwise, it is zero.

Analysis and Evaluation:

Balance of Workload: The jobs are distributed to ensure an even distribution of workload among the groups, minimizing the maximum completion time (makespan).

Minimized Makespan: The evolutionary model aims to minimize the makespan by ensuring that the total processing time in each group does not exceed the due date constraints.

Job Lateness: All jobs are scheduled such that the job lateness is zero, indicating that all jobs are completed on or before their respective due dates.

Flow Time Optimization: The flow times are optimized to ensure the sequential order of job processing within each group, reducing idle time and maximizing efficiency.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N
		Order	Processing Time (hours)	Due Date (-th hour)			Group Setting	Order	Processing Time (Hours)	Flow Time (Hours)		Due Date (-th hours)	Job Lateness	
2														
3		1	13	25		Group 1	G1	18	6	6	<=	29	0	
4		2	9	24			G1	20	8	14	<=	33	0	
5		3	5	15			G1	8	7	21	<=	26	0	
6		4	15	36		Group 2	G2	13	8	8	<=	23	0	
7		5	7	22			G2	10	7	15	<=	17	0	
8		6	8	18			G2	5	7	22	<=	22	0	
9		7	8	17		Group 3	G3	7	8	8	<=	17	0	
10		8	7	26			G3	2	9	17	<=	24	0	
11		9	4	20			G3	4	5	22	<=	36	0	
12		10	7	17		Group 4	G4	3	5	5	<=	15	0	
13		11	11	29			G4	14	3	8	<=	10	0	
14		12	5	14			G4	19	12	20	<=	34	0	
15		13	8	23		Group 5	G5	9	4	4	<=	20	0	
16		14	3	10			G5	12	5	9	<=	14	0	
17		15	12	25			G5	1	13	22	<=	25	0	

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Solver Parameters

Set Objective:

To: ☐ Max ☒ Min ☐ Value Of:

By Changing Variable Cells:

Subject to the Constraints:

-
-
-
-
-

☐ Make Unconstrained Variables Non-Negative

Select a Solving Method:

Solving Method
Select the GRG Nonlinear engine for Solver Problems that are smooth nonlinear. Select the LP Simplex engine for linear Solver Problems, and select the Evolutionary engine for Solver problems that are non-smooth.

Options

All Methods | GRG Nonlinear | **Evolutionary**

Convergence:

Mutation Rate:

Population Size:

Random Seed:

Maximum Time without improvement:

☒ Require Bounds on Variables

(b) 4 workers in a group, 5 groups of workers.

Objective:

Minimize the makespan, the total time required to complete all jobs.

Constraints:

- Each job must be assigned to one group.
- Total processing time per group should be less than or equal to the available working hours.
- Jobs within each group are to be scheduled to meet their respective due dates as closely as possible to avoid lateness.

Generated schedule and analysis:

This schedule was generated to balance the workload across 5 groups while aiming to minimize the makespan. The flow times, due dates, and job lateness are calculated to assess the efficiency of the schedule.

Group 1

Order	Processing Time (hours)	Flow Time (hours)	Due Date (-th hours)	Job Lateness
9	4	4	20	0
16	2	6	8	0
1	13	19	25	0
4	15	34	36	0

Group 2

Order	Processing Time (hours)	Flow Time (hours)	Due Date (-th hours)	Job Lateness
7	8	8	17	0
3	5	13	15	0
15	12	25	25	0
20	8	33	33	0

Group 3

Order	Processing Time (hours)	Flow Time (hours)	Due Date (-th hours)	Job Lateness
12	5	5	14	0
2	9	14	24	0
13	8	22	23	0
18	6	28	29	0

Group 4

Order	Processing Time (hours)	Flow Time (hours)	Due Date (-th hours)	Job Lateness
14	3	3	10	0
10	7	10	17	0
6	8	18	18	0
11	11	29	29	0

Group 5

Order	Processing Time (hours)	Flow Time (hours)	Due Date (-th hours)	Job Lateness
17	7	7	16	0
5	7	14	22	0
8	7	21	26	0
19	12	33	34	0

Total Processing time:

The total processing time for each group is 34 hours.

Parameters:

Group Configuration:

Number of Groups: 5 groups

Workers per Group: 4 workers

Analysis and Evaluation:

The workload is fairly distributed among the 5 groups.

Minimized Makespan:

The largest flow time observed is 34 hours, achieved by Group 1.

All jobs meet their respective due dates with zero lateness, except for one job in Group 1 which meets its due date exactly.

Flow Time Efficiency:

The flow time values indicate the total hours taken from the start of the group's first job to the completion of its last job.

[illegible]

Solver Parameters

Set Objective:

To: ☐ Max ☒ Min ☐ Value Of:

By Changing Variable Cells:

Subject to the Constraints:

-
-
-
-

☐ Make Unconstrained Variables Non-Negative

Select a Solving Method:

Solving Method

Select the GRG Nonlinear engine for Solver Problems that are smooth nonlinear. Select the LP Simplex engine for linear Solver Problems, and select the Evolutionary engine for Solver problems that are non-smooth.

Options

☒ All Methods ☐ GRG Nonlinear ☐ Evolutionary

Constraint Precision:

☒ Use Automatic Scaling

☐ Show Iteration Results

Solving with Integer Constraints

☐ Ignore Integer Constraints

Integer Optimality (%):

Solving Limits

Max Time (Seconds):

Iterations:

Evolutionary and Integer Constraints

Max Subproblems:

Max Feasible Solutions:

5 (c) 5 workers in a group, 4 groups of workers.

Objective:

Minimize the makespan, which is the total time required to complete all jobs.

Constraints:

Each job must be assigned to exactly one group.

The total processing time for each group should be balanced to ensure the workload is evenly distributed.

Job lateness should be minimized.

Generated schedule and analysis:

The table below shows the order and processing times for the 20 jobs distributed across the 4 groups before optimization using the solver. The columns include:

Group 1

Order	Processing Time (hours)	Flow Time (hours)	Due Date (-th hour)	Job Lateness
G1	6	8	≤ 18	0
G1	7	16	≤ 17	0
G1	3	21	≤ 15	1
G1	4	36	≤ 36	0
G1	8	43	≤ 26	1

Group 2

Order	Processing Time (hours)	Flow Time (hours)	Due Date (-th hour)	Job Lateness
G2	5	7	≤ 22	0
G2	9	11	≤ 20	0
G2	14	14	≤ 10	1
G2	2	23	≤ 24	0
G2	18	29	≤ 29	0

Group 3

Order	Processing Time (hours)	Flow Time (hours)	Due Date (-th hour)	Job Lateness
G3	13	8	≤ 23	0
G3	10	15	≤ 17	0
G3	11	26	≤ 29	0

G3	20	34	≤ 33	1
G3	15	46	≤ 25	1

Group 4

Order	Processing Time (hours)	Flow Time (hours)	Due Date (-th hour)	Job Lateness
G4	12	5	≤ 14	0
G4	16	7	≤ 8	0
G4	17	14	≤ 16	0
G4	1	27	≤ 25	1
G4	19	39	≤ 34	1

Parameters :

Order: The job number.

Processing Time (hours): The time required to complete the job.

Due Date (-th hour): The deadline by which the job should be completed.

Flow Time (hours): The cumulative time from the start until the completion of a particular job in a group.

Job Lateness: Indicates if the job is completed on time or late (0 if on time, 1 if late).

Analysis and evaluation:

Total Flow Time: The total flow time for each group is calculated to evaluate the workload distribution.

Job Lateness: Jobs with lateness indicate delays in meeting due dates. Reducing lateness is critical for optimizing the schedule.

Makespan: The makespan is determined by the group with the maximum flow time. The goal is to minimize this value to ensure efficient scheduling.

The first screenshot shows the **Solver Parameters** dialog box. The **Set Objective:** is $\$H\22 . The **To:** radio buttons are set to **Min**. The **By Changing Variable Cells:** is $\$H\$2:\$H\21 . The **Subject to the Constraints:** list includes:

- $\$H\$2:\$H\$21 \leq 20$
- $\$H\$2:\$H\$21 = \text{AllDifferent}$
- $\$H\$2:\$H\$21 = \text{Integer}$
- $\$H\$2:\$H\$21 \geq 1$
- $\$I\$2:\$I\$21 \leq \$L\$2:\$L\21

The **Select a Solving Method:** dropdown is set to **Evolutionary**. The **Solving Method** section states: "Select the GRG Nonlinear engine for Solver Problems that are smooth nonlinear. Select the LP Simplex engine for linear Solver Problems, and select the Evolutionary engine for Solver problems that are non-smooth."

The second screenshot shows the **Solver Options** dialog box. The **Convergence:** is set to 0.0001. The **Mutation Rate:** is 0.075. The **Population Size:** is 250. The **Random Seed:** is 0. The **Maximum Time without improvement:** is 30. The **Require Bounds on Variables** checkbox is checked.

The background spreadsheet shows a table with columns: **Order**, **Processing Time (hours)**, **Due Date (-th hour)**, **Flow time (hours)**, and **Job Lateness**. The **Flow time (hours)** column is highlighted in blue, and the **Job Lateness** column is highlighted in green. The **Flow time (hours)** column contains values: 8, 16, 21, 36, 43, 7, 11, 14, 23, 29, 8, 15, 26, 34, 46, 5, 7, 14, 27, 39. The **Job Lateness** column contains values: 0, 0, 1, 0, 1, 0, 0, 1, 0, 0, 0, 0, 0, 1, 1, 0, 0, 0, 1, 1.

Conclusion:

By analysing a,b and c we can say 7 groups are the best solution for Ali's workshop as it will take less flow time 22 hours.

